CS 457 - Homework Assignment 1: Data Types

Due Date: Monday, September 04 at 11:59 pm

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Purpose:
Demonstrate your understanding of data types by examining a non-trivial public dataset and identifying the NOIR analytical data types of each of the data attributes (variables). This skill set will be used frequently in future data analytics.

Part 1 (30 points)

Question
List each attribute (column) given in the dataset, choose its analytical data type (NOIR type) and explain why you consider this type. Answer without explanation will not be accepted.

Files:
List of all earthquakes for past month:
https://earthquake.usgs.gov/earthquakes/feed/v1.0/summary/all_month.csv

Resources:
About earthquakes and dataset details:
https://earthquake.usgs.gov/earthquakes/feed/v1.0/csv.php

Explanation:

Time: This is clearly an interval type as it is a continuous numerical measurement with a well defined interval. Also, Zero time does not imply absence of the measured characteristics.
Latitude: Same as Above. -ve data can easily be used as well.
Longitude: Same as Above. -ve data can easily be used as well.
Depth: Although earthquakes have extremely low intensity when they are at depth 0 aka the surface level. This mean that it can only be felt through special instruments. However, using depth 0 as a measure for the absence of tectonic activity is a bad idea as magnitude perhaps would be more suited for that. In the context of depth being an absolute length measurement and nothing else, it being zero should not signify absence of anything. In otherwords, depth will always exist in this data and zero depth is also a valid parameter. Since, depth 0 has a clear definition of it being at the surface level. We can think of it as a ratio type. In this type, having a clear definition for zero value is must.

Mag: Ritcher Scale does not have an absolute zero point at zero magnitude. The Richter scale is a logarithmic scale, and it measures the amplitude of seismic waves. I would disagree with what's written in slides that it's a ratio scale. It is not a true ratio scale because it lacks a true zero point where there is an absence of earthquake activity. Theoretically, there is no lower limit to the scale. Small earthquakes have positive Richter scale values, and larger earthquakes have larger positive values, while very small earthquakes have negative values. So, it's not a ratio scale because you can't make meaningful statements about ratios of earthquake magnitudes that easily. The website (https://earthquake.usgs.gov/data/comcat/index.php#mag) has defined it as starting from zero and zero implying absence and going all the way to 100. However, the dataset (all_month.csv) opposes this notion as there were few entities having negative magnitude. (Basically hard to define an absolute zero point). Still I cannot label it as interval because there is no theoretical limit to what it can be. So the best option is to make assumptions as zero being absolute absence and higher magnitudes being higher intensities. This generalization is not accurate but should suffice for mainstream purposes. So Ratio it is.
magType: Ordinal classification as they have a discrete rank and can be ordered (Types have ranges assigned to them based on that)
nst: Number of stations involved in reporting. This should be ratio as zero should imply absence of any reporting. However, in our dataset there is always some magnitude which means it must be measured in some station. Yet there were instances when there was empty field. Assuming empty fields to be zero and assuming this signifies absence of reporting stations. Then this should be ratio.
Gap: This is azimuthal gap between adjacent stations. Since this is a derived numerical quantity from NST, we can also think of it as ratio and zero gap implying zero nst.
Dmin: Another derived numerical quantity from NST. Zero Dmin implies zero nst.
Rms: Ratio as the dataset contains zero values which actually imply absence.
Net: It should be nominal as its just a qualitative measure with no possible operations. Not even ordering is possible.
Id: Same as above.
Updated: Interval as its based on time which is also an interval. 0 values in it does not signify absence. The intervals are also easy to define.
Place: Nominal as the data is simply qualitative with no well defined order. We can try to order it based on place name and distance, but how do we decide which compass direction to label as the greater one? So in otherwords its quite tricky to define an order. So Nominal.
Type: Nominal as its simply a qualitative measure.

horizontalError: Ratio as it's a numerical value with possible operations and order. However, practically speaking, its hard to imagine an errorless measurement. So its hard to think of error being possibly zero in any case. Dataset also does not contain the value '0'. However, if it were to exist it would have a well defined definition and that would be 'Absence of error'. So ratio it is.
depthError: Same as above.
magError : Same as above.
magNst: Ratio (Same as NST).
Status: Nominal as status is a qualitative measurement with no possible operations and with no well defined order.
locationSource: Same as above.
magSource: Same as Above.

Part 2 (30 points)

Question
Pick analytical data type (NOIR type) for each of the items below and explain why you consider this type. Answer without explanation will not be accepted.

Explanation:

1. Time with possible values AM or PM.

Ordinal. While AM and PM are qualitative measures, they still have a well-defined order to them. We know that PM>AM. As the time of a day starts with AM. End > Start. Therefore, relational operations should also be possible on AM/PM Time.

2. Brightness as measured by a light meter.

Ratio as 0 brightness has well defined definition of absence of light. Numerical operations can be performed and has a well defined order.

3. Brightness as measured by people's judgments.

Nominal. People's judgement can never be quantitative and hence its qualitative. So we can assign labels/categories to brightness but we can not define a rank/order for it. For example how do we know which adjective describes a brighter light: "bright' and 'radiant'. Sure we can add degrees of adjectives but those degrees would only be valid for a particular adjective and not for all possible adjectives (categories).

4. Angles as measured in degrees between 0^o and 360^o.

Interval as its well defined from 0 to 360, and has a range of continuous numerical values. The value 0 also does not denote absence.

5. Bronze, Silver and Gold medals as awarded at the Olympics.

Ordinal as all three are qualitative but their order is well defined making relational operations possible. Gold > Silver > Bronze.

6. Height above sea level.

Clearly ratio as zero denotes absence. Numerical operations are possible, and values can be ordered.

7. Number of patients in a hospital.

Ratio as there is a true zero point which signifies that the hospital is empty. Numerical operations should also be possible.

8. ISBN numbers for books. (Look up the format on the Web):

Nominal; The numbers are qualitative and therefore no numerical operation should be possible. However, there is no well defined order for them. Nor does it makes sense to rank them numerically as bigger ISBN does not mean a latest book.

9. Ability to pass light in terms of the following values: opaque, translucent, transparent.

Ordinal: The three choices are qualitative measurements however a well defined order should be easy to specify transparent > translucent > opaque. This should make relational operations possible.

10. Military rank.

Ordinal: The ranks are qualitative and their order is already defined.

11. Distance from the center of HU campus.

Ratio, as the data is numerical and should have numerical operations, be well ordered. We also have a well defined true zero point which is the center itself.

12. Density of a substance in grams per cubic centimeter.

Ratio: Density has a true zero point which could be thought of as absence of mass or absence of volume. Numerical operations should be possible and its values should be easy to order.

Part 3 (40 points)

Question

Assume you are doing a study of Habib University students' academic and demographic characteristics and storing this information as a dataset. Identify total of 12 attributes (3N, 3O, 3I, and 3R) and explain why you think is the correct data type. Answer without explanation will not be accepted.

For example: *Academic Year: (Freshman, Sophomore, Junior, Senior) = Ordinal* (because they can be ordered based on their admission year)

Explanation:

3N:

Name | ID | Major

These are qualitative and have no well defined order to it. (Assume that ID is generated randomly and not chronologically). These are simply categories that define a person.

3O:

Batch: Numerical operations should not be possible but relational operations should be possible on this category as Class of 2025 > Class of 2024. Newer batch can be defined as greater in rank.

Academic Standing: Easy to establish rank here as well. Good Academic Standing > Academic Probation.

Scholarship: Scholarships are discrete categories at Habib. But they can be ranked easily. Yohsin > Excellence > Merit. Such relational operations should prove useful in evaluating potential performance of a student.

3I:
CGPA: We can think of this interval as [0:4]. It also lacks a true zero point as 0 CGPA does not mean absence of academic performance. It could very well mean that this is the starting semester of someone.
DOB: Date of birth is a well-defined interval and has certain date time interval. There is no true zero point here. Relational operations should be possible.
Avg Time Per Course: Since Habib University takes course evaluations in which students do a self evaluation and write their hours committed to each course. We can take avg time per course from that to evaluate the effort put by a student. This attribute will be of interval scale as zero time will not imply absence of time.

3R:
Credits Currently Taking: 0 here would be a true zero point and would imply withdrawal from semester.
Number of Clubs: 0 here would imply a true zero point, meaning that student is part of no clubs at all.
Number of Sports: 0 here would imply that student plays no sport at all. In all of these ratio scales, numerical operations, and relational operations should be possible making it easier to evaluate current performance.

Deliverable: Submit a PDF format of your homework solution on Canvas.